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ASAK v2 – RINGKASAN TEKNIS

Aksara Sasak Digital | Game-Ified Learning Platform

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STACK & PLATFORM

Mobile : Android (Kotlin) + Android Studio
AR : Unity + Vuforia (image marker), model .fbx dari Blender
AI : CNN (handwriting recognition) + Voice Recognition (sudah jadi di v1)
AI Tutor : Gemini API (gemini-1.5-flash) – system prompt injection, bukan fine-tuning
Backend : REST API (PHP/Node, belum spesifik), MySQL
Local : Room DB (offline cache)
HTTP : Retrofit + OkHttp
Infra : Cron job (streak reset 23:59 harian, leaderboard reset Senin 00:00)

DATA – 18 HURUF (Aksara Baluq Olas)

HA, NA, CA, RA, KA, DA, TA, SA, WA, LA, MA, GA, BA, PA, JA, YA, NYA, NGA

Tiap huruf menyimpan:

- kode, nama_latin, unicode_char, tipe_fonem
- mnemonic (deskripsi visual hafalan)
- langkah_tulis (JSON array step-by-step)
- audio_url, model_3d_url, barcode_image_url
- level_unlock (huruf terbuka di level berapa)
- konfusable_with (JSON ID huruf yang sering tertukar – data distraktor)

Distribusi level:

- L1: HA, NA, CA, RA, KA
- L2: DA, TA, SA, WA, LA
- L3: MA, GA, BA, PA
- L4: JA, YA, NYA, NGA
- L5-L6: review, ujian besar

QUIZ – 5 TIPE

- Tipe 1 Gambar → pilih nama (4 opsi, 3 distraktor algoritmik)
Tipe 2 Audio → pilih gambar aksara (replay ke-2 kena -1 diamond)
Tipe 3 Lihat aksara → tulis di canvas → CNN nilai akurasi, $\geq 75\%$ = benar [L2+]
Tipe 4 Lihat aksara → ucapkan → voice recognition, $\geq 70\%$ = benar [L2+]
Tipe 5 Drag & drop pasangkan 4 aksara ↔ 4 nama, partial score 5XP/pasang [L3+]
Bonus Speed Round (L5+): 10 soal, 60 detik, benar +6 detik

Scoring per soal:

Benar < 5 detik : +10 XP + 5 XP speed bonus
Benar normal : +10 XP
Salah : 0 XP + masuk retry queue (max 3 soal diulang di akhir)

Hint : -2 diamond
Retry queue : max 3 soal salah diulang di akhir sesi

ALGORITMA ADAPTIVE QUIZ

Input : user_id, jumlah_soal
Output : list soal ter-shuffle

1. Ambil history akurasi 30 hari terakhir per huruf
2. Kategorikan:
 - lemah = akurasi < 60% → 40% soal
 - sedang = 60-79% → 30% soal
 - kuat = >= 80% → 20% soal
 - belum = belum pernah → 10% soal
3. Untuk tiap huruf, tentukan tipe soal optimal dari riwayat kesalahan
4. Pilih distraktor dari konfusable_with (bukan acak)
5. Urutkan: mudah → sedang → susah (scaffolded)

Pasangan sering keliru (dari data survei 105 siswa):
DA↔TA, GA↔JA, NYA↔NGA, WA↔MA, BA↔PA

ALGORITMA SPACED REPETITION (SM-2 disederhanakan)

Score >= 90 : mastery +15, interval = mastery/10 hari
Score 70-89 : mastery +5, interval = max(1, mastery/15) hari
Score < 70 : mastery -10, interval = 1 hari (review besok)

Status huruf:
mastery >= 90 → 'mastered'
mastery >= 50 → 'practiced'
mastery < 50 → 'introduced'

Disimpan di: user_huruf_progress.next_review_at

ALGORITMA XP AWARD

xp_final = ROUND(base_xp * multiplier)

Multiplier streak:
streak >= 30 : x2.0
streak >= 14 : x1.5
streak >= 7 : x1.25
streak >= 3 : x1.1

Item Double XP aktif : multiplier x2.0 (stack)
Top 10 liga leaderboard : multiplier x1.2 (stack)

XP base per aktivitas:

Selesai pelajaran	: 50-100 XP
Quiz benar	: 10 XP (+5 speed bonus)
Review harian	: 20 XP
Task harian all done	: +50 XP bonus
Lulus level	: reward_xp per tabel levels

ALGORITMA STREAK (Cron 23:59 harian)

Untuk tiap user:

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Jika daily_tasks.all_done == TRUE:
    Jika last_active == kemarin → streak +1
    Jika last_active == hari ini → skip (sudah dihitung)
    Selain itu (terputus):
        Jika streak_shield_active → streak +1, shield = FALSE
        Selain itu → streak = 1 (reset)
    Jika all_done == FALSE → tidak ubah streak, besok otomatis terputus
```

streak_shield = item Diamond yang bisa dipakai untuk menyelamatkan 1 hari skip

ALGORITMA LEADERBOARD RESET (Cron Senin 00:00)

1. Snapshot ranking tiap liga berdasarkan weekly_xp
2. Award hadiah:
 - Rank 1 : +30 diamond + double XP 2 jam + badge Raja Minggu
 - Rank 2 : +20 diamond + double XP 1 jam
 - Rank 3 : +10 diamond + double XP 30 mnt
 - Top 4-10 : +5 diamond + +20% XP 30 mnt
3. Promotion/demotion:
 - Top 20% liga → naik satu liga
 - Bottom 20% liga → turun satu liga
4. Reset weekly_xp semua user = 0
5. Push notifikasi hasil liga ke semua user

Liga tier: bronze → silver → gold → diamond → gem

ALGORITMA ACHIEVEMENT CHECK

Dipanggil setiap ada event: quiz_selesai, streak_update, ar_scan, level_selesai, artikel_dibaca, dll.

Saat achievement terpenuhi:

- Catat ke user_achievements
- award_xp(ach.xp_reward)
- award_diamond(ach.diamond_reward)
- Jika ada → unlock AR budaya / profile frame
- Tandai notifikasi (popup muncul sesi berikutnya)

Contoh trigger achievement:

STREAK_3/7/14/30/100 : streak milestone
PERFECT_QUIZ : score 100% satu sesi
COMBO_10 : 10 benar berturut dalam 1 sesi
LULUS_UJIAN_1 : lulus ujian besar level 3
LULUS_UJIAN_FINAL : lulus ujian besar level 6
KOLEKTOR_AR_3D : scan semua 18 huruf AR
SCAN_PERTAMA : scan AR pertama kali

DATABASE – TABEL UTAMA

users

id, username, email, password_hash, display_name, avatar_url,
total_xp, weekly_xp, diamond_balance,
current_streak, longest_streak, last_active_date,
league_tier ENUM(bronze/silver/gold/diamond/gem),
role ENUM(user/admin)

aksara_huruf

id, kode, nama_latin, unicode_char, tipe_fonem,
mnemonik, langkah_tulis JSON, audio_url,
model_3d_url, barcode_image_url, level_unlock,
konfusable_with JSON, sejarah_singkat

user_huruf_progress

user_id FK, huruf_id FK,
status ENUM(locked/introduced/practiced/mastered),
total_attempts, correct_count,
write_accuracy_avg, voice_accuracy_avg,
ar_scanned BOOL, last_practiced, next_review_at, mastery_score

levels

id (1-6), nama, deskripsi, huruf_diperkenalkan JSON,
xp_target, pelajaran_count, ada_ujian_besar BOOL,
reward_xp, reward_diamond

pelajaran

id, level_id FK, urutan, nama,
tipe ENUM(intro/write/voice/ar/review/quiz_mini/ujian_besar/budaya),
konten_json JSON, xp_reward, passing_score (default 70)

user_level_progress

user_id FK, level_id FK,
status ENUM(locked/in_progress/completed/mastered),
current_pelajaran, xp_earned, started_at, completed_at

quiz_sessions

user_id FK, pelajaran_id FK (NULL=free practice),
tipe_quiz ENUM(mini/besar/daily/free),
total_soal, benar, salah, score_pct, xp_earned,
diamond_used, duration_seconds,
answers_json JSON [{huruf_id, tipe_soal, user_answer, correct,
time_ms}]

```

achievements
  id, kode UNIQUE, nama, deskripsi, ikon_url,
  xp_reward, diamond_reward,
  unlock_ar_id FK (NULL), unlock_frame_id FK (NULL),
  kategori ENUM(belajar/streak/ar/budaya/quiz/sosial)

user_achievements
  user_id FK, achievement_id FK, earned_at, notified BOOL

diamond_transactions
  user_id FK, tipe ENUM(earn/spend),
  jumlah, sumber VARCHAR, saldo_sesudah, created_at

daily_tasks
  user_id FK, tanggal DATE,
  task_review_done BOOL, task_quiz_done BOOL, task_practice_done BOOL,
  all_done BOOL, xp_bonus_claimed BOOL, diamond_bonus_claimed BOOL

ar_budaya
  id, nama, deskripsi, model_3d_url, marker_image_url, thumbnail_url,
  unlock_type ENUM(default/achievement/diamond/exam),
  diamond_cost (default 15),
  unlock_achievement_id FK (NULL),
  has_interactive BOOL, interactive_diamond_cost (default 20)

user_ar_unlocks
  user_id FK, ar_budaya_id FK,
  unlock_method ENUM(achievement/diamond/exam),
  interactive_unlocked BOOL, unlocked_at

leaderboard_weekly
  user_id FK, week_start DATE,
  weekly_xp, rank, league_tier, reward_claimed BOOL

```

AR SYSTEM

AR Huruf (18 barcode):

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Engine      : Unity + Vuforia
Trigger     : Scan barcode di buku fisik
Konten      : Model 3D huruf, animasi stroke, audio, info nama/bunyi
Premium     : Animasi interaktif (-20 diamond)
Event       : Kirim ar_scan_logged ke backend setelah scan
Backend     : Update user_huruf_progress.ar_scanned, cek achievement

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AR Budaya (6 objek Lombok):

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Gendang Beleg      → default (selesai L2)
Perisaian          → lulus Ujian Besar 1 / 15 diamond
Masjid Bayan Beleg → streak 14 hari / 15 diamond
Bale Lumbung       → achievement Sejarah / 15 diamond
Taman Narmada      → streak 30 hari / 15 diamond
Circuit Mandalika  → achievement Penjelajah / 15 diamond
Semua unlock       → lulus Ujian Final otomatis

```

Alur AR teknis:

Kamera → Vuforia scan 30ms/frame → cocok image target →
Load .fbx → render overlay → kirim event ke backend

AI TUTOR — GEMINI API

Model : gemini-1.5-flash (bukan fine-tuning, bukan pretraining)
Metode : System Prompt Context Injection per sesi

Cara kerja:

Setiap sesi chat, kirim ke Gemini:
- system: konteks ASAK (18 huruf, level, gamifikasi, fitur)
- contents: history chat (max 10 pesan terakhir) + pesan baru

Kenapa tidak fine-tuning:

- Tidak butuh data training berlabel
- Tidak butuh GPU / biaya training
- Update konten = edit string, bukan retrain
- Free tier Gemini: 1 juta token/bulan, cukup untuk MVP

Aturan di system prompt:

- Jawab bahasa Indonesia, ramah, sesuai SD-SMP
- Fokus: aksara Sasak, cara belajar, fitur ASAK, budaya Lombok
- Jangan sebut diri "Gemini", gunakan "ASAK AI Tutor"

Keamanan API key:

Gunakan backend proxy (tidak hardcode di APK)
Client Android → backend kamu → Gemini API

Android implementation:

AiTutorViewModel.kt : StateFlow<List<ChatMessage>>, sendMessage(),
buildGeminiRequest()
GeminiApiService.kt : Retrofit wrapper ke Gemini REST endpoint
fragment_ai_tutor.xml : RecyclerView bubble chat + EditText + chip
suggestion

MOBILE — PEMBAGIAN SCREEN (Kotlin/Android Studio)

RENDI (AI)

Screen 1: AiTutorFragment
- RecyclerView bubble chat, TypingIndicator, chip suggestion
- State: empty, loading, error
Screen 2: QuizTulisFragment (Tipe 3)
- Canvas input, integrasi model handwriting CNN
- Feedback animasi benar/salah, XP trigger
Screen 3: QuizSuaraFragment (Tipe 4)
- Mic input, integrasi voice recognition model
Teknis: GeminiApiService, AiTutorViewModel

DESTI (AR)

Screen 1: ArScanFragment

- UI kamera + overlay viewfinder
- Animasi scanning, transisi ke Unity AR player
- Info panel overlay (nama, bunyi, cara baca)

Screen 2: ArBudayaFragment

- Galeri konten budaya (lock/unlock grid)
- Tap → AR viewer 3D, progress "X/18 ditemukan"

Teknis: Unity-Android bridge (UnityPlayerActivity / AAR), ARCore fallback

RIDHO (Backend/DB)

Screen 1: LeaderboardFragment

- Tab: liga sekarang | riwayat minggu lalu
- Header: nama liga + countdown reset
- List ranking, row user sendiri sticky

Screen 2: ProfilStatistikFragment

- Total XP, sesi, akurasi rata-rata
- Progress bar per huruf, grafik XP 7 hari

Teknis: REST API endpoint (XP/streak/achievement/leaderboard), cron jobs, Room DB local cache, Repository layer + Retrofit service

KANDA (Frontend)

Screen 1: HomeFragment (Beranda)

- Header: XP bar, diamond balance, streak counter
- Daily task card, level progress, quick action buttons
- Koordinasi konten dari semua modul

Screen 2: AchievementFragment

- Grid terkunci/terbuka, filter, detail popup
- Animasi particle saat achievement unlock

Screen 3: OnboardingActivity (3-4 slide)

Teknis: Design System (warna, tipografi, komponen reusable), animasi transisi shared element, XP gain animation, achievement popup confetti

ROADMAP (14 Minggu)

Sprint 1 (M1-3) : Migrasi DB, seed 18 huruf, UI Beranda, XP engine, Quiz Tipe 1&2

Sprint 2 (M4-6) : Diamond economy, streak cron, achievement engine, Quiz Tipe 3&4

Sprint 3 (M7-9) : Level 1-6 full, Ujian Besar, spaced repetition, adaptive quiz, sertifikat PDF

Sprint 4 (M10-12): AR integrasi Unity, AR Budaya unlock, leaderboard liga, double XP, push notif

Sprint 5 (M13-14): Black box testing, USE questionnaire, optimasi APK, offline mode, admin dashboard

v3 scope: sandangan (tanda vokal), gabungan huruf jadi kata, angka Sasak, iOS, multiplayer duel

RISIKO TEKNIS

- Akurasi CNN handwriting awal <80%, butuh augmentasi data lebih
- APK bisa >100MB (Unity + 3D model + audio), perlu lazy load asset
- AR boros baterai dan panas di HP low-end
- Cron job streak + leaderboard butuh edge case handling ketat
- API key Gemini harus lewat backend proxy, jangan expose di APK
- Fitur AI (voice/write) butuh internet stabil – daerah terpencil jadi hambatan
- Scope v2 hanya isolated character, bukan kata/kalimat bermakna

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